As a pathologist, I studied disease through a microscope. As an epidemiologist, I studied patterns across communities.

Then it became personal.

After I lost my job, my family lost our health insurance. The next month, my husband was bitten by a tick and became ill with Lyme disease.

For immigrants like my husband, language can make it harder to understand tests, discuss bills, and explain symptoms.

I'm Yuwei Zhang. That experience led me to build PathwayAI.

The burden starts before a diagnosis. It includes uncertainty, finding care, missed work, and help from family. Patients need a next step. Communities need to understand what is missing.

Here is a fictional example. A person records two hundred twenty-five dollars in care and travel payments, three missed workdays, and eight hours of unpaid help.

These stay separate. Unknown information stays unknown. Repeated records could show how the burden changes over time.

At community level, we compare Dutchess and Orange counties using public data and published research. Federal sources include CDC surveillance, Census, HRSA, and CDC PLACES. New York surveillance adds local context.

County cost figures are illustrations based on study averages, not measured county spending or personal bills.

Now let me show you what people can do.

Before a bite, find prevention information. After a bite, find removal steps and time-sensitive advice. A clinician decides whether preventive treatment is appropriate.

If you lack insurance, start with community health centers and questions about income-based fees. Find coverage and hospital financial-assistance resources. Care is not automatically free, but there is a place to start.

Next, prepare for a visit. Our fictional patient describes a tick encounter, symptoms two weeks later, and a blood test they report was positive.

PathwayAI organizes these into a timeline. Symptom onset and time to diagnosis stay separate. The person can review, correct, and download the summary. The clinician still needs the original test report.

People can write in their preferred language. With permission, optional AI organizes and translates an English summary beside the original. People review both before sharing. It does not replace a qualified medical interpreter. The basic English organizer also works without external AI.

AI helps organize words. Arithmetic calculates costs. The patient remains at the center.

Patients can also prepare questions about bills or transport eligibility, cost, and booking.

For county and community teams, the planning brief connects a barrier with a possible response, responsible partners, funding needs, and follow-up. Did someone reach care? Did support arrive? The app cannot provide rides, approve assistance, or create funding.

Our main planning users are county and community teams. Patients and caregivers use the preparation tools directly.

People living with Lyme disease and a county health commissioner reviewed the working app. Feedback highlighted confusing choices, reading difficulty, and the need for practical next steps. It has guided improvements and our next round of simpler flows.

Sixty-four automated tests passed. These check software behavior. They do not establish clinical accuracy, better access, or savings. Those outcomes need evaluation.

The pilot is live on Streamlit. Next, I plan a small pilot with willing patient groups, health centers, and care navigators. We will test usefulness, verify resources, and follow support connections.

Free patient access supported by grants and community partners is the proposed model. Funding is not secured. Shared patient-report collection stays off until consent and secure storage are ready.

Longer-term records are the next research goal. We are seeking partners to help.

PathwayAI: Full Economic Burden. Making the hidden patient journey visible, and actionable.

Thank you to the patients and experts helping shape it.